1 Title

House Price Prediction using Machine Learning Model Comparison

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Course: **AI & ML**
Task: **Model Comparison**
Date: **(today's date)**

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2 Introduction

This project aims to predict house prices using machine learning algorithms. The California Housing dataset is used to train and evaluate multiple regression models. The objective of this task is to compare different models and determine which model performs best for predicting house prices.

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3 Dataset Description

The California Housing dataset contains information collected from housing districts in California. The dataset includes features such as median income, house age, average rooms, population, and geographic location.

The target variable is **median house value**, which represents the price of houses in each district.

You can also mention:

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Total Samples: 20640
Number of Features: 8
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4 Methodology

The following steps were performed in this project:

1. Load the California Housing dataset
2. Perform data preprocessing and feature scaling
3. Split the dataset into training (80%) and testing (20%) sets
4. Train multiple regression models
5. Evaluate models using RMSE and R^2 score
6. Compare model performance

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5 Models Used

• Linear Regression
• Ridge Regression
• Decision Tree Regressor

Explain shortly:

Linear Regression is a basic regression model used as a baseline.
Ridge Regression is a regularized version of linear regression that reduces overfitting.
Decision Tree Regressor captures non-linear relationships in the data.

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6 Evaluation Metrics

RMSE (Root Mean Squared Error)
Measures the difference between predicted and actual values.

R^2 Score

Indicates how well the model explains the variance in the dataset.

7 Results

Model	RMSE	R^2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

Based on the evaluation metrics, the Decision Tree model achieved the lowest RMSE and the highest R^2 score. Therefore, it performed better compared to the other models.

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8 Conclusion

In this project, multiple machine learning models were implemented to predict house prices. After comparing the performance of the models, the Decision Tree Regressor showed better prediction performance based on RMSE and R^2 score. This experiment highlights the importance of model comparison and proper evaluation when building machine learning models.